

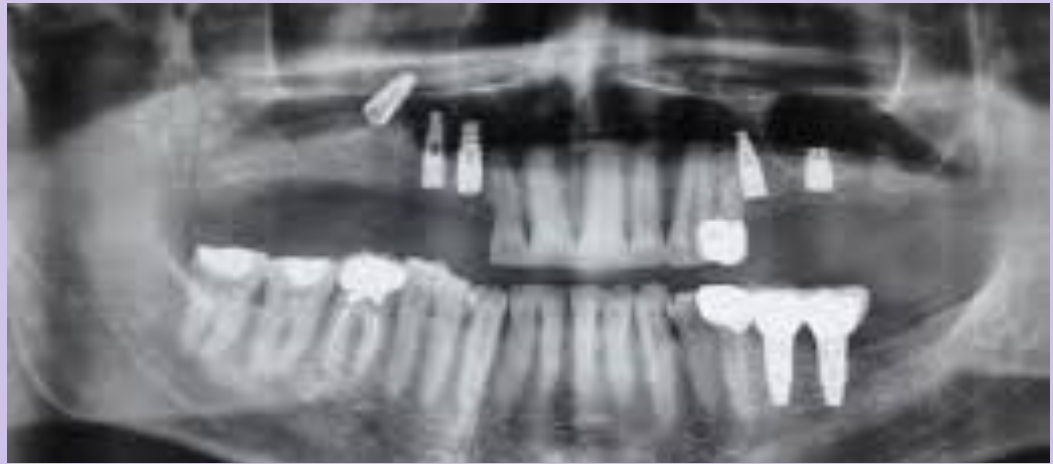


Problem and Solution

Dental implants are one of the top 3 reasons to sue a dentist for **malpractice**. Prosthodontic specialty, where implant surgeries are a part of, had the highest number of lawsuits (32.4%), caused by **complications** such as nerve damage, sinus perforation infection, and bone damage. Moreover, a single tooth implant costs **\$3,000 to \$5,000** on average, which can be expensive for those with weak financial supports. Our solution utilizes **IMU sensors** to track the exact angulation of the dental drill in real time during the procedure. It will increase the **precision** of the surgery and **decrease the cost** of operation.



Osseonews "Implant Placed in Inferior Alveolar Canal: Timeline for Patient Recovery?" OsseoNews, 30 Jan. 2012
www.osseonews.com/implant-placed-in-inferior-alveolar-canal-timeline-for-patient-recovery/.



History and Comparison to Alternatives

Traditionally, **surgical dental guides** are used to ensure proper angulation. These are essentially molds of a patient's teeth with a designated hole for the dental drill, ensuring it drills into the patients tissue at the correct angle. However, these guides can compress on the adjacent soft tissue, which can cause it to deform. **Poor soft tissue management** can alter the angulation of the guide or make it shift position. Additionally, there may be errors of **up to 5° and 2.3 mm even with modern guides**.

These guides range from \$250-\$400 per piece, not including shipping, and are usually not covered by insurance. Our design is far cheaper by comparison, as it is reusable patient-to-patient, and is more accurate as it tracks the real time angulation of the dental drill during the procedure.

In order to ensure maximum accuracy in real time, our team chose to utilize **BNO055 sensors** that utilize **wifi** to transmit data. Other real time surgical devices, such as the **Navident EVO**, utilize computer vision or other more expensive methods to calculate angulation and assist with other implant procedures. However this can become extremely expensive, with the Navident EVO selling for \$40,320. Our device is not only cheaper, but simpler to use, and can be used in educational settings as well to help train dental students.

Concept Selection Matrix of Prototypes

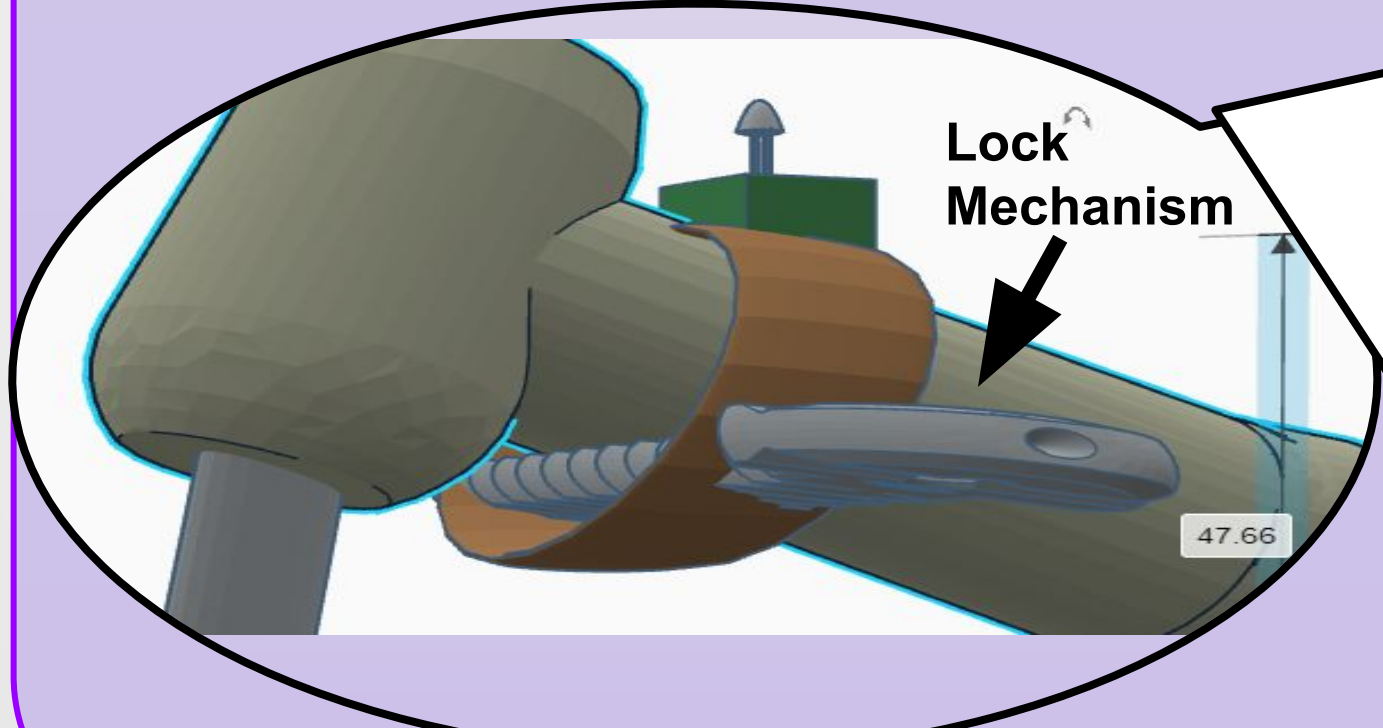
(1 - 5)	Wired + MPU6050	Bluetooth + BNO055	WiFi + BNO055
Accuracy & Precision	3	5	5
Real-time Response	5	3	4
User Interface	3	5	5
Cost Efficiency	5	3	4
Sterilization & Reusability	4	5	5
Portability & Compactness	1	5	5
Maintenance & Calibration	3	5	5
Total	24	31	33

Detailed Solution

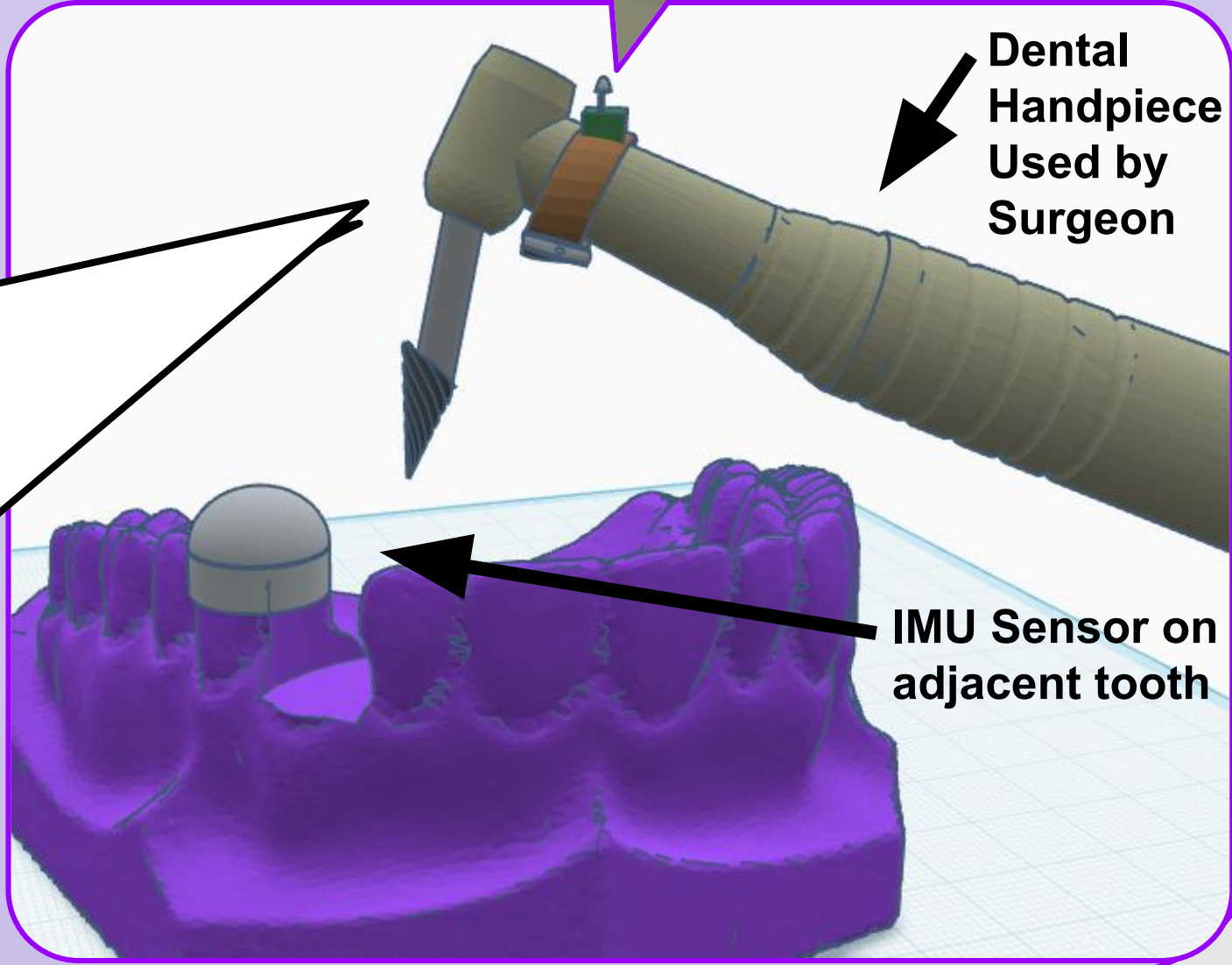
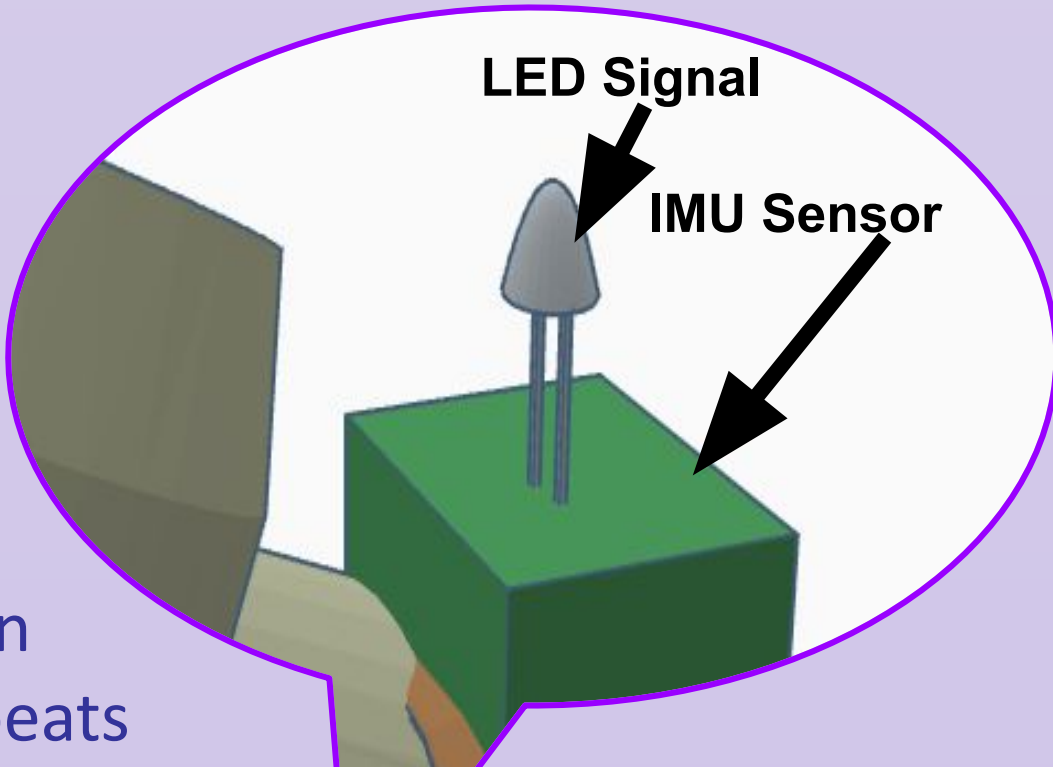
Our solution is to add an attachment equipped with IMU sensors to a dental implant drill and an adjacent tooth to prevent dental implant complications related to improper angulation of the drill. Our current prototype uses an accelerometer and gyroscope to calculate the orientation of two different setpoints: one for the anchoring sensor, which will be placed on the adjacent tooth, and the other attached to the dental drill used for the procedure. The **microcontroller** responsible for transmitting the angular data to the computer will also be included in the unit, producing an ideal dimension of less than 5x5 mm per part.

Process: After the surgeon sets the desired orientation of the handpiece, the relative orientation between the two sensors is saved and later used as a benchmark. During the surgery, the angular values of the drill are read in real-time; if there is a deviation between the benchmark and the current relative angle, an external LED is turned on to alert the surgeons.

Computation: The IMU sensors get their accelerometer/gyroscope values and the final roll/pitch values using a 96:4 complementary filter. Then, they calculate the orientation between the sensors and store it as the ideal objective. The patient's jaw is not being fixed during the surgery, so a sensor is required to be the benchmark and acquire the orientation of the handpiece. Throughout the surgery, the device repeats to calculate the orientation and check if it's heading towards the objective within the set range. If the handpiece orientation exceeds this range, the LED activates, alerting the surgeon.

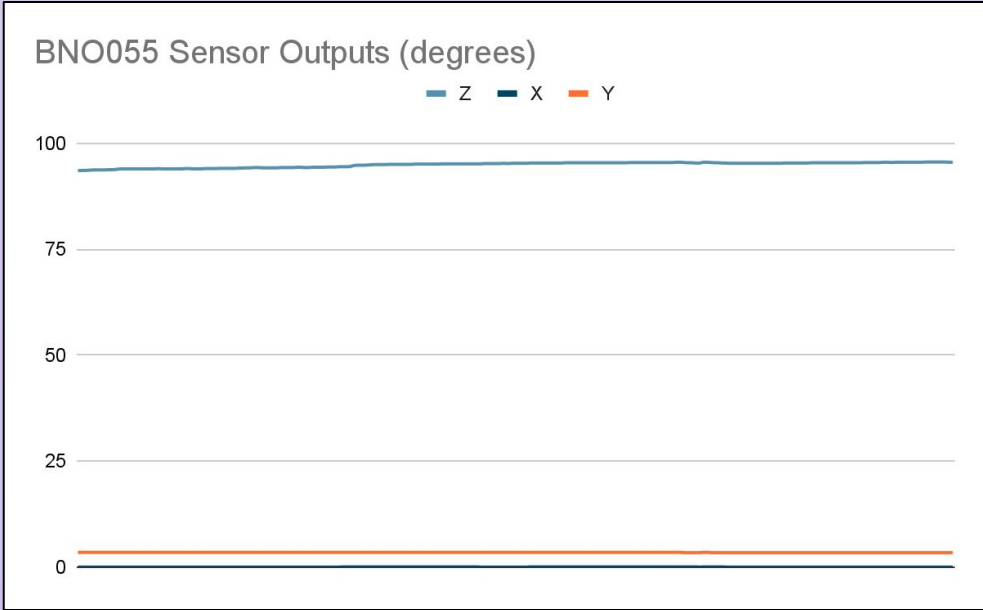


Images generated using TinkerCAD



Testing and Benchmarking

Our device is classified as a **Class I Dental Handpiece Accessory** by the FDA, so it only needs to meet testing benchmarks for accuracy, precision, and biocompatibility of materials. Since gyroscopic sensors invoke error over time, an observation of the data stream is necessary, also known as **Zero Point Drift Test**. Sensors are put at various angles in a stationary state for 30 minutes and the collected data can be used to evaluate on the sensor's **stability**. From previous research by Yoon Nam, the device was tested through human applications, where dental residents simulated simple operations with and without the device. In order to make a numerical evaluation, the standard deviation of the data stream can be calculated to represent the deviations in the outputs into a single number: a large standard of deviation would imply instability of the sensor and vice versa. The Zero Point Drift Test shows an outstanding result for the BNO055 as it shows in the graph: it returned a steady stream of data for 30 minutes.



Ethical Challenges

The development and implementation of an IMU-guided dental implant surgery device creates many ethical considerations that must be examined. The **ADA (American Dental Association)** clearly states in their code of ethics that any marketing claim must be backed with accepted scientific research. Claiming that our device “Ensures precise angulation” is acceptable with rigorous testing and reproducible data showing that the IMU sensors accurately calculate angulation. However, claiming that this device “reduces surgical errors” is unethical without human testing showing that implant errors were reduced. Marketing must also be clear that this device is not strictly necessary during dental implant procedures, it simply enhances the outcome, and other methods are available.

In terms of **accessibility**, many insurance companies view dental implants as a cosmetic procedure, limiting coverage options, while some insurance companies cover up to 50% of the procedure. This means that our device would likely not be covered by most insurance companies. However, our device eliminates the need and the cost of dental guides all together. It is cheaper to the dental office, thus reducing the price of the procedure overall and making it more accessible. The device should be clearly positioned as a **supplemental educational tool** that enhances drilling technique training, without implying it confers specialist credentials or substitutes accredited clinical training.

Technical Challenges

The main technical challenge involves acquiring **accurate angular data** while maintaining **feasibility** for dental professionals. For example, a device with wired connection would stream data at an extremely **low latency**, but may cause visual interference during the procedure. Similarly, wireless modules must support real time data streaming so that the surgeons can receive outputs with minimal delay.

The device must also easily **integrate** into the various tools of a dental surgeon as well as the arrangement of their operating room. Thus, the IMU sensors will contain adjustable clasps modules allowing for **universal attachment**. Finally, the device must have **power** throughout the operation. A wired connection will supply sufficient power, although this may impede on a surgeon's control of the dental drill, similar to the wired data streaming method. A rechargeable device with **long-lasting battery life** will solve this.

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